CORNELL TECH | XR Collaboratory

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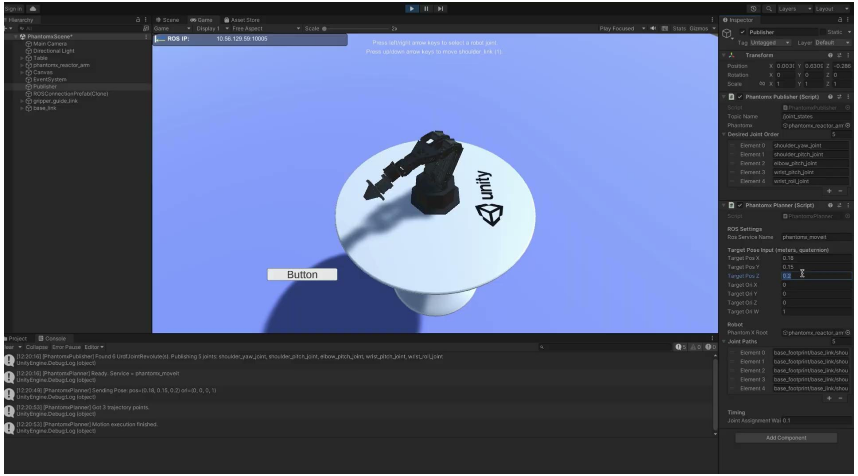
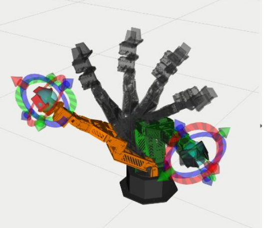
ROS 2 Control & MoveIt Integration for PhantomX Reactor Arm with Unity-based Spatial Interaction for VR/AR

Abstract

The project develops a complete ROS 2 control package for the PhantomX Reactor robotic arm, integrating ros2_control for hardware-level actuation and MoveIt 2 for motion planning.
Building on this foundation, a Unity-based interface is introduced to provide intuitive spatial interaction in VR/AR environments. Users can point, select, or tap virtual objects, and the robot automatically moves to the desired pose via the ROS2–Unity bridge.
This work establishes a modular and extensible manipulation pipeline suitable for robotics research, immersive interfaces, and human–robot interaction applications.





Package Features

- **ROS 2 Control:** URDF hardware interface with joint state and trajectory control.
- **MoveIt 2:** IK + OMPL planning with RViz visualization.
- **Unity:** Gestures (In Progress) → target poses → real-time robot motion.

Package Repository: